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EDUCATION

Irvington High School **Fremont, CA**
Computer Science + Math

- Outside Coursework: Multivariable Calculus, Linear Alegbra and Diff Eq, Intro/Intermediate C++
- Coursework: AP World History, AP Computer Science A, AP Macroeconomics, AP English Language, AP US History, AP Physics 1, AP Calculus BC, AP Precalculus, AP Statistics, AP Government and Politics, AP Physics C: Mechanics, AP Physics C: E&M, AP Chemistry

RESEARCH EXPERIENCE

Research Internship
Stanford Artificial Intelligence Lab *May 2026 - Present*

- Worked under Dr. Chen Shani under the Stanford Natural Language Processing Group.
- Designed and implemented a fine-tuning pipeline comparing aggregated (majority-vote) vs. non-aggregated (per-annotator) label training strategies across two datasets: a vaccine hesitancy stance classification dataset and ChaosNLI (an NLI dataset with 100 annotations per example).
- Fine-tuned and evaluated 10 transformer models spanning encoder (BERT-base, RoBERTa-base, ModernBERT, NeoBERT, DeBERTa-v3-large) and decoder (GPT-2, SmolLM2-135M, SmolLM2-360M, Pythia-160M, GPT-X2-125M) architectures.
- Implemented 5-fold stratified cross-validation for robust evaluation, with per-fold random annotator sampling for ChaosNLI to maximize use of its annotation richness.
- Identified training instability in RoBERTa under non-aggregated labels (macro F1 std dev ±0.21 vs. ±0.03 for all other configs), and found that non-aggregated training did not consistently improve performance for either encoder or decoder models.
- Built a modular codebase with a shared utility library covering data loading, CV loop, dataset classes, and metrics, supporting both datasets and all models from a single interface.

Student Researcher
MIT Criticial Data *July 2026 - Present*

- Co-designed a pre-registered, factorial prompt experiment auditing how LLMs disclose the provenance of medical knowledge they generate, with independent verification of every returned citation.
- Built a codebook from scratch to classify AI-generated citations as fabricated, real-with-bibliographic-errors, or fully accurate, iterating the category rules against 10-15 pilot citations before freezing it.
- Ran a 4-tier escalating prompt protocol (plain question → sourcing request → provenance interrogation → adversarial pushback) across multiple LLMs on adolescent mental health questions, logging model version, run date, and retrieval setting for every session.
- Manually verified all returned citations against PubMed, Crossref, and Google Scholar to compute fabrication and bibliographic-error rates, without AI assistance in the coding process.
- Conducted independent double-coding with a teammate and computed inter-rater reliability (Cohen's kappa), with a third teammate adjudicating disagreements below the reliability threshold.

Regression Research Lead

UCLA Mathematics *December 2025 - June 2026*

- Conducted research under PhD Candidate William Legg, focusing on mathematical concepts such as Linear Algebra and Data Science.
  Developed and implemented OLS, Ridge, and Lasso regression models from first principles using NumPy to predict building thermal efficiency.
- Derived the Normal Equation for closed-form solutions and implemented Coordinate Descent for Lasso regularization to handle feature collinearity.
- Achieved an accuracy of 90.8%, matching industry-standard scikit-learn benchmarks while ensuring 100% mathematical transparency in the codebase.
- Analyzed a factorial design dataset of 768 building configurations, evaluating the impact of 8 geometric features on primary energy consumption.

PROJECTS

Lyze: Multimodal RAG Intelligence System *November 2025 - Present*

- Built a multimodal RAG pipeline using FastAPI and Qdrant, enabling semantic search across PDFs, images, audio, and video in a unified 768-dimensional vector store.
- Engineered a multimodal ingestion system using LLaVA for image captioning and Whisper for audio/video transcription, supporting cross-modal querying across 4 file types in a single search.
- Designed a provider-agnostic architecture supporting 4 AI backends (Ollama, OpenAI, Anthropic, Gemini) with a React + TypeScript frontend and Python REST API.
- Garnered over 7k views across platforms around the world in the United States, India, United Kingdom, and more.

LEADERSHIP

Competitive Programming Club **Irvington High School, CA**

President *August 2023 - Present*

- Managed large-scale educational outreach by prepping 50+ elementary students for the American Computer Science League, focusing on foundational logic and coding proficiency.
- Spearheaded technical training sessions for club members, specializing in algorithmic problem-solving and optimization techniques tailored for USACO (USA Computing Olympiad) and USAAIO (USA AI Olympiad) competitions.
- Developed instructional curriculum covering complex data structures and algorithms, including Breadth-First Search (BFS), Depth-First Search (DFS), and Dynamic Programming.

Math Club **Irvington High School, CA**

Vice President *August 2023 - Present*

- Execute organizational operations by managing meeting agendas, official records, and logistical planning to ensure seamless club administration for 100+ members.
- Facilitate advanced technical seminars on complex topics including Geometry, Number Theory, and Combinatorics, simplifying high-level concepts for student engagement.
- Coordinated school-wide participation in prestigious regional and national tournaments, including BMT (Berkeley), SMT (Stanford), ASMA, and Purple Comet.

SKILLS

- **Programming**: C++, Java, Python, Typescript, HTML, MATLAB
- **Technical**: Machine Learning, Linear Regression, Data Analyst, Language Interpretability,
- **Frameworks/Libraries**: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

AWARDS

- **IAI²O 2026 Silver Medalist (USAAIO)** - One of 59 students nationally selected to represent the US internationally (Science Track), earned Silver Medal (19/60).
- **2025 AP Scholar Award** - Gained a 5 on 3+ AP Collegeboard Exams
- **SMT 2025 Distinguished Honorable Mention -** Top 20% in Team + Power Round
- **BMT 2025 Honorable Mention** - Top 50% in Algebra Round
- **SMT 2026 Distinguished Honorable Mention** - Top 20% in Team + Power Round